Sunwhi Kim, Ph.D.

김선휘 · also published as Sunwhi Kimm

Assistant Professor, Department of Bio-Healthcare
Hwasung Medi-Science University

skim@hsmu.ac.kr   gdrpaul3@gmail.com   office 031-369-9100

ORCID   GOOGLE SCHOLAR   GITHUB   LAB WEBSITE   WEB CV   CV SOURCE   YOUTUBE

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PROFESSIONAL APPOINTMENTS

2025–present

Assistant Professor
Department of Bio-Healthcare, Hwasung Medi-Science University, Republic of Korea

- Department Chair (학과장) since September 2025; principal investigator of the Neural Systems Laboratory.
- Research spanning neural-circuit mechanisms of social and defensive behaviour and the cognitive human factors of human–AI interaction.

Apr 2019–Dec 2024

Postdoctoral Fellow
Department of Anatomy & Neurobiology, College of Medicine, University of Tennessee Health Science Center, Memphis, TN, USA

Advisor: Dr. Il Hwan Kim

- Circuit- and network-specific dissection of the mechanisms underlying social dysfunction in Shank3 models of autism spectrum disorder.
- Developed the split-Cre circuit-targeting toolkit (Addgene #187614, #187615) and the Monosynaptically-interconnected Network Module (MNM) analysis method.

EDUCATION

Mar 2013–Aug 2019

Ph.D. in Biopsychology (이학박사, Doctor of Science)
Department of Psychology, Korea University

Advisor: Dr. June-Seek Choi · Dissertation: The functions of the amygdala sub-nuclei and prelimbic cortex in an approach-avoidance conflict situation

Mar 2006–Feb 2013

B.A. in Psychology
College of Liberal Arts, Department of Psychology, Korea University

`Mar 2006–Feb 2013` **B.Eng. in Brain and Cognitive Sciences (interdisciplinary major)**
College of Informatics, Korea University

CITATION METRICS

162	6	4
CITATIONS	H-INDEX	I10-INDEX

Google Scholar, as of 2026-08-24

PEER-REVIEWED JOURNAL ARTICLES

*Published under both "Kim" and "Kimm". Each work is listed under the surname used in the original publication. * equal contribution; † corresponding author.*

1. Kim YE, Kim M*, **Kim S***, Lee R*, Ujihara Y, Marquez-Wilkins EM, Jiang YH, Yang E, Kim H, Lee C, Park C†, Kim IH† (2025). Endothelial SHANK3 regulates tight junctions in the neonatal mouse blood-brain barrier through β-Catenin signaling. *Nature Communications 16, 1407*. `doi:10.1038/s41467-025-56720-1`

2. **Kimm S**, Kim JJ, Choi JS† (2023). The central amygdala modulates distinctive conflict-like behaviors in a naturalistic foraging task. *Frontiers in Behavioral Neuroscience 17, 1212884.* `doi:10.3389/fnbeh.2023.1212884`

3. **Kim S**, Kim YE, Song I, Ujihara Y, Kim N, Jiang YH, Yin HH, Lee TH, Kim IH† (2022). Neural circuit pathology driven by Shank3 mutation disrupts social behaviors. *Cell Reports 39(10), 110906.* `doi:10.1016/j.celrep.2022.110906` COVER ARTICLE

4. **Kim S***, Kim YE*, Kim IH† (2022). Simultaneous analysis of social behaviors and neural responses in mice using round social arena system. *STAR Protocols 3(4), 101722.* `doi:10.1016/j.xpro.2022.101722`

5. Kim YE*, **Kim S***, Kim IH† (2022). Neural circuit-specific gene manipulation in mouse brain in vivo using split-intein-mediated split-Cre system. *STAR Protocols 3(4), 101807.* `doi:10.1016/j.xpro.2022.101807`

6. Kim IH†, Kim N, **Kim S**, Toda K, Catavero CM, Courtland JL, Yin HH, Soderling S† (2020). Dysregulation of the synaptic cytoskeleton in the PFC drives neural circuit pathology, leading to social dysfunction. *Cell Reports 32(4), 107965.* `doi:10.1016/j.celrep.2020.107965`

7. **Kimm S**, Choi JS† (2018). Sensory and motivational modulation of immediate and delayed defensive responses under dynamic threat. *Journal of Neuroscience Methods 307, 84-94.* `doi:10.1016/j.jneumeth.2018.06.023`

8. Lee JH, **Kimm S**, Han JS, Choi JS† (2018). Chasing as a model of psychogenic stress: characterization of physiological and behavioral responses. *Stress 21(4), 323-332*. `doi:10.1080/10253890.2018.1455090`

PREPRINTS & MANUSCRIPTS

1. **Kim S**, Kim SY (2026). Human factors in detecting AI-generated portraits: age, sex, device, and confidence. *arXiv, arXiv:2603.24048*. `doi:10.48550/arXiv.2603.24048` PREPRINT

2. **Kim S†**, Kim YE, Ujihara Y, Kim IH† (2024). Monosynaptically-interconnected network module (MNM) approach for high-resolution brain sub-network analysis. *bioRxiv, 2024.02.19.581007*. `doi:10.1101/2024.02.19.581007` PREPRINT

CONFERENCE PROCEEDINGS

1. Jeong JH, **Kimm S**, Lee J, Choi JS† (2019). Differential encoding of events during approach–avoidance conflict situation by the prelimbic cortex. *IBRO Reports 6, S313-S314*.

2. Choi JS†, Lee JH, **Kim S**, Han JS (2017). Validating an animal model of stress using a chasing robot. *Proceedings of the 15th Conference of the Japanese Society for Cognitive Psychology (日本認知心理学会発表論文集)*.

3. **Kim S**, Hong EH, Jie HS, Jeong JH, Choi JS† (2015). Control and data acquisition of rodent experiments using Arduino-based system. *Korean Association for Laboratory Animal Science International Symposium (한국실험동물학회 학술발표대회 논문집), 124*.

RESEARCH FUNDING

`2025–2026.02 (year 1 of 5)` **Silver digital literacy education and ESG campaign (industry–academia regional cooperation)**
Gyeonggi-do RISE Programme, HSMU RISE Centre · Role: Curriculum and courseware development lead

`2025.04–2026.02` **Long-term neurobehavioral safety assessment of transcranial focused ultrasound neuromodulation**
HSMU New Faculty Research Grant · Role: Principal Investigator · ₩5,000,000

GRANTS UNDER REVIEW

`2026.04–2029.12` **Development of AI learning frameworks based on brain cognitive science and biological learning principles**
Ministry of Science and ICT / Institute of Information & Communications Technology Planning & Evaluation (IITP)

`2026–2028 (3 years, proposed)` **AI-based quantification of social behaviors in group-housed rodents and their whole-brain neural correlates**
National Research Foundation of Korea (NRF)

TEACHING

TERM	COURSE	CODE	CR.	INSTITUTION
2026-1	**Cognitive-Behavioral Experiment and Design** 인지행동실험과설계	443525	3	Hwasung Medi-Science University
2026-1	**Introduction to Neuroscience** 뇌과학개론	500387	3	Hwasung Medi-Science University
2026-1	**Introduction to Preclinical Studies** 전임상개론	500386	3	Hwasung Medi-Science University
2025-2	**Understanding Genetic Engineering** 유전공학의이해	500500	3	Hwasung Medi-Science University
2025-2	**Animal Behavior Experiment** 동물행동실험	500501	3	Hwasung Medi-Science University
2025-2	**Basic Animal Experiment** 기초동물실험	500498	3	Hwasung Medi-Science University
2025-1	**Introduction to Neuroscience** 뇌과학개론	500387	3	Hwasung Medi-Science University
2025-1	**Introduction to Preclinical Studies** 전임상개론	500386	3	Hwasung Medi-Science University
2025-1	**General Biology** 일반생물학	500384	3	Hwasung Medi-Science University
2025-1	**Physiology** 생리학	242519	2	Hwasung Medi-Science University (Dept. of Nursing)
2024 Spring	**Guest lecture: “Autism spectrum disorder” — Developmental and Molecular Neurobiology**	ANAT 825		University of Tennessee Health Science Center
2022 Spring	**Guest lecture: “Autism spectrum disorder and neural circuit-based approach” — Developmental and Molecular Neurobiology**	ANAT 825		University of Tennessee Health Science Center
2016-1	**Workshop lecture: animal tracking and behavioral measurement with ANY-maze** 행동신경과학 워크샵 — ANY-maze 사용 및 활용 방법			Korea University

INVITED TALKS & GUEST LECTURES

2025

“3D 프린터로 취미에서 창업까지: 아이디어를 현실로 만드는 기술 (From hobby to start-up with 3D printing)”
HSMU ‘CareerFit’ programme (invited campus lecture) · Hwasung Medi-Science University, Korea

2025

“뇌 신경 회로 타겟팅 기술을 이용한 자폐 스펙트럼 장애 메커니즘 연구 (Circuit-targeting approaches to autism spectrum disorder)”
1st HSMU Colloquium, Center for Teaching and Learning · Hwasung Medi-Science University, Korea

2022

“Neural circuit-selective Shank3 mutation disrupts social behaviors”
Memphis Korean Scientists Association · Memphis, TN, USA

2022

“Neural circuit-selective Shank3 mutation disrupts social behaviors”
35th US-Korea Conference (UKC) · Washington, D.C., USA

2022

“Neural circuit pathology driven by Shank3 mutation disrupts social behaviors”
Neuroscience Institute Spring Seminar, University of Tennessee Health Science Center · Memphis, TN, USA

2021

“Neural circuit pathology driven by Shank3 mutation disrupts social behaviors”
BK21 Seminar, Korea University · Seoul, Republic of Korea

CONFERENCE PRESENTATIONS

2024

“The miniature neural network module (MNM) approach for mapping and analyzing modular subnetworks in the brain”
Kim S, Kim YE, Ujihara Y, Kim IH · Cold Spring Harbor Laboratory Neuronal Circuits Meeting · Cold Spring Harbor, NY, USA

2018

“Rat’s head-withdrawal reflex depends on visual and vibrissal information and on auditory conditioning”
Kim S, Lee JY, Choi JS · Korean Society of Cognitive and Biological Psychology · Korea

2016

“A modified version of Lima-Dill model predicts approach-avoidance behavior of rats facing a predator-like robot”
Kim S, Jie HS, Choi JS · Society for Neuroscience (Neuroscience 2016) · San Diego, CA, USA

2016

“Amygdala regulates approach-avoidance conflict under predatory threat in rats: a model-based approach”
Kim S, Choi JS · NeuroSplash (Data Blitz) · Korea

2015 **"Control and data acquisition of animal behavior experiments using Arduino-based system"**
Kim S, Choi JS · Korean Psychological Association Annual Conference · Seoul, Korea

2015 **"Control and data acquisition of rodent experiments using Arduino-based system"**
Kim S, Hong EW, Jie HS, Jeong JH, Lee JH, Choi JS · Korean Association for Laboratory Animal Science International Symposium · Incheon, Korea

2015 **"Central and basolateral amygdala lesions disrupt risk avoidance but spare reflexive defensive system"**
Kimm S, Choi JS · Korean Society for Brain and Neural Sciences · Korea

2015 **"A cost-effective system for experimental control and data acquisition using Arduino"**
Kimm S, Hong EH, Jie HS, Choi JS · Korean Society of Cognitive and Biological Psychology · Jeju, Korea

2014 **"Amygdala modulates approach-avoidance but not reflexive withdrawal in a semi-naturalistic conflict situation"**
Kimm SW, Kim DY, Choi JS · Annual Pavlovian Society Meeting · Seattle, WA, USA

2014 **"Amygdala modulates approach-avoidance but not reflexive withdrawal in semi-naturalistic conflict situation"**
Kimm S, Kim D, Choi JS · Korean Society for Brain and Neural Sciences · Seoul, Korea

2014 **"Partial amygdala lesions increase approach to the robot predator and impede the expression of overall defensive behaviors except for reflexive withdrawal"**
Kimm S, Kim D, Choi JS · NeuroSplash · Korea

2014 **"Search of sign stimulus for rats' defensive behavior using a cat-like robot"**
Kimm S, Lee YK, Choi JS · Korean Society of Cognitive and Biological Psychology · Buyeo, Korea

2013 **"Past experience of mild threat reduces anxiety in rats during dynamic interaction with a robot stressor"**
Kimm S, Choi JS · 16th Annual International Meeting of the Korean Society for Brain and Neural Sciences · Korea University, Seoul, Korea

2013 **"Lobsterbot, a possible new tool for the study of animal's risk evaluation and decision making"**
Kimm S, Lee JY, Choi JS · Korean Society for Brain and Cognitive Science Conference · Seoul National University, Seoul, Korea

2013 **"Fear of predation: artificial predation-like motions by robotic agents and their effects on defensive behavior"**
Kimm S, Choi JS · NeuroSplash · Korea

SOFTWARE & REGISTERED PROGRAMS

YEAR	PROGRAM	REGISTRATION	DESCRIPTION
2016	**Small Animal Defensive Response Measurement and Analysis Program**	C-2016-025109	Classical-conditioning experiment control and analysis software
2015	**ActiMon — 24-hour small-animal activity monitoring system**	C-2015-008076	Continuous 24-hour rodent activity monitoring
2015	**Serial Reaction Time Task**	C-2015-008009	Attention and working-memory test battery for rodents

Open-source research and teaching tools

YEAR	PROJECT	DESCRIPTION
2025	**Real vs. AI — public portrait-discrimination web experiment**	Two-alternative forced-choice web experiment used to collect the 1,664-participant dataset behind the 2026 AI-portrait study
2026	**hsmu_ai_detection_public**	Open stimulus set, metadata and de-identified trial-level behavioural data for the AI-generated portrait detection study
2022	**Addgene plasmid deposition: split-Cre toolkit**	pAAV-Ef1a-CreN-InteinN (#187614) and pAAV-Ef1a-InteinC-CreC (#187615) — circuit-specific gene-manipulation reagents distributed to the community

MENTORING & ADVISING

2025– **11 undergraduate advisees, Dept. of Bio-Healthcare**
Faculty academic advisor, Hwasung Medi-Science University

2021–2024 **Yusuke Ujihara, B.A.**
Graduate student, UTHSC

2024.05 **Elizabeth Rachel Dang**
M.D. candidate, UTHSC

2014 **Teaching assistant, Introduction to Brain Science (ISC218)**
Korea University International Summer Campus

2013	**Teaching assistant, Learning Psychology** Korea University, Dept. of Psychology

INSTITUTIONAL & PROFESSIONAL SERVICE

2025.09–2027.08	**Department Chair, Department of Bio-Healthcare** 화성의과학대학교 / Hwasung Medi-Science University
2026.02	**Expert content-validity reviewer, institutional core-competency scale development** 화성의과학대학교
2025.10–2025.12	**Member, Scholarship Review Committee** 화성의과학대학교
2025	**Reviewer, Liberal 101 scholarship programme** 화성의과학대학교
2025.06	**Member, Major Curriculum Subcommittee** 화성의과학대학교
2025–	**Faculty academic advisor (11 advisees), academic-probation counselling, merit-scholarship nomination** 화성의과학대학교 바이오헬스케어학과
2025–	**Departmental website build, admissions presentations, chairing faculty meetings, curriculum and timetable planning** 화성의과학대학교 바이오헬스케어학과
2015–	**Member — Korean Psychological Association; Korean Association for Laboratory Animal Science**

AWARDS & HONORS

2022	**Journal cover selection, Cell Reports 39(10)** Cell Press
2018	**Research Scholarship** Korea University
2015	**BK21 PLUS Scholarship** Korea University
2014	**Institute of Foreign Language Studies Work Scholarship** Korea University

2014 **BK21 PLUS Scholarship**
Korea University

2013 **Research Assistant Scholarship**
Korea University

2013 **BK21 PLUS Scholarship**
Korea University

2010 **Work Scholarship**
Korea University

2006 **Freshman Special Scholarship**
Korea University

CERTIFICATIONS & TRAINING

2026-02-25 **Research Ethics for Principal Investigators (Science & Engineering)**
국가과학기술인력개발원 (KIRD)

2025-07-19 **Laboratory Animal Technician Practical Workshop (theory module)**
한국실험동물학회 인증위원회 (KALAS)

2025-05-24 **Laboratory Safety Education & Training for new researchers (8 h)**
화성의과학대학교

2025-04-30 **Online Training for Human Subjects Researchers on the Ethical Conduct of Research**
국가생명윤리정책원 (KoNIBP)

2019-02-28 **Biosafety Training / Laboratory Practicum Safety Training**
한국생명공학연구원 국가연구안전관리본부 (KRIBB)

2018-09-04 **Biosafety Training**
한국생명공학연구원 국가연구안전관리본부 (KRIBB)

2013-05-09 **CITI Program — Human Research, Basic Course**
Collaborative Institutional Training Initiative (via Korea University)

MEDIA COVERAGE

2022 **Neural circuit pathology driven by Shank3 mutation disrupts social behaviors — 한빛사 인터뷰**
한빛사 (BRIC)

2022 **Autism-linked mutation disrupts brain circuit to change social behavior**
Spectrum News

OUTREACH

2026–	**Makers — student maker & technology club** Faculty advisor. Student club for AI coding, generative media, 3D printing, Arduino and automation, run as a bridge between the lab's human–AI research and hands-on student prototyping.
2025	**Public participant-recruitment campaign, AI-portrait perception study** Campus and online science-engagement campaign that recruited 1,843 public participants for the AI-generated portrait detection experiment.
2025	**화성왭성 AI creative contest — video entry “꿈속의 화성”** Generative-AI music video produced as team lead; an applied companion to the lab's generative-media research.

PEER REVIEW

2024, 2025	**STAR Protocols (Cell Press)** Four review reports, including both rounds of one 2025 manuscript
2022	**Genes, Brain and Behavior**
2022	**Scientific Reports**
2014	**Behavioural Brain Research (Elsevier)** two manuscripts, co-review with doctoral advisor

TECHNICAL SKILLS

AREA	DETAIL
Neural Circuit Methods	Viral circuit tracing (retrograde / anterograde / monosynaptic rabies), Split-Cre and intersectional circuit-specific gene manipulation, Optogenetics, In vivo calcium imaging (fiber photometry, miniscope), In vivo electrophysiology, Immunohistochemistry and whole-brain mapping
Behaviour	Rodent social, defensive and approach-avoidance paradigms, Semi-naturalistic robot-predator foraging tasks, Fear conditioning, extinction and context renewal, Automated tracking (DeepLabCut, ANY-maze, SAM-based multi-animal tracking)
Human Experimentation	Large-scale online psychophysics (2AFC, reaction-time modelling), IRB protocol design and human-subjects research governance, Signal detection theory
Computational	Python, R, MATLAB, Mixed-effects and Bayesian modelling, RNA-seq / differential-expression and network analysis, Machine-learning behavioural classification

AREA	DETAIL
Engineering	Arduino-based experiment control and data acquisition, 3D printing and rapid prototyping, Web application development (Firebase, GitHub Pages) for experiment delivery
Languages	Korean (native), English (professional working proficiency)

REFERENCES

Dr. June-Seek Choi
Professor of Psychology · Korea University · j-schoi@korea.ac.kr

Dr. Il Hwan Kim
Associate Professor, Dept. of Anatomy & Neurobiology · University of Tennessee Health Science Center · ikim9@uthsc.edu